

Exercise 1

SQL Fundamentals: SELECT and filtering

1. SELECT *
FROM employees;

id	first_name	last_name	department	salary	hire-date	city
1.	Alice	Green	IT	70 000	2000-01-10	JHB
2.	Brian	Lee	HR	45 000	2019-08-22	CPT
3.	Cathy	Zulu	Finance	65 000	2018-07-18	IDB
4.	David	Makana	Marketing	50 000	2021-11-05	PTA
5.	Eva	Naiabo	IT	72 000	2017-09-30	JHB

2. SELECT DISTINCT department
FROM employee;

department
IT
HR
Finance
Marketing

3. SELECT first_name,
last_name
FROM employees
ORDER BY salary DESC;

first_name	last_name
Eva	Naiabo
Alice	Green
Cathy	Zulu
David	Makana
Brian	Lee

4. SELECT id, first-name, last-name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

id	first-name	last-name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

5. SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'IT';

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. SELECT id, first-name, last-name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65 000

7. SELECT id, first-name, last-name, department
FROM employee
WHERE department = 'HR' OR department = 'Marketing';

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. SELECT id, first-name, last-name, department
FROM employee
WHERE department

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. SELECT id, first-name, last-name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SELECT id, first-name, last-name, department, salary,
city
FROM employees

WHERE department = 'IT' AND salary > 65 000
AND city = JOHANNESBURG

id	firstname	last-name	department	salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg